

BSIMAR: An Autoregressive Compact Model for FinFETs

Anonymous CVPR submission

Paper ID 20525

Abstract

As process nodes advance, the behavior of semiconductor devices spans weak inversion to saturation, with pronounced short-channel, mobility, quantization, self-heating, and non-quasi-static (NQS) effects. Traditional compact models (e.g., BSIM-CMG) deliver physics-grounded equations but require large, coupled parameter sets, intensive calibration, and brittle transfer across geometries and technologies. Recently, neural network approaches have improved fit and speed, yet most train separate I - V / Q - V / C - V models, struggle with device-aware conditioning, charge continuity, and data imbalance, and generalize poorly to unseen nodes or new devices.

We introduce BSIMAR, an autoregressive compact model that treats device characterization as next-token prediction and jointly generates I - V , Q - V , and C - V with internal consistency. BSIMAR uses imbalance-aware losses and scales with data and model size, enabling efficient fine-tuning across devices/technologies. The model runs on GPUs for accelerated transistor-level simulation and integrates cleanly with SPICE-like flows. Experiments show improved accuracy, tail-region fidelity, and cross-device generalization relative to NN baselines while reducing calibration effort. Code and models will be released on GitHub.

1. Introduction

Complementary metal-oxide-semiconductor (CMOS) logic and fin field-effect transistors (FinFETs) underpin modern integrated circuits. As technology nodes shrink, planar metal-oxide-semiconductor field-effect transistors (MOSFETs) give way to multi-gate FinFETs, which improve electrostatic control and leakage, enabling continued gains in performance and energy efficiency (Figure 1).

A compact model is a reduced-order, physics-informed equation set that maps terminal voltages, temperature, geometry, and process parameters to device currents, charges, and capacitances for efficient circuit-level simulation. In practice, compact models run in SPICE-like simulators

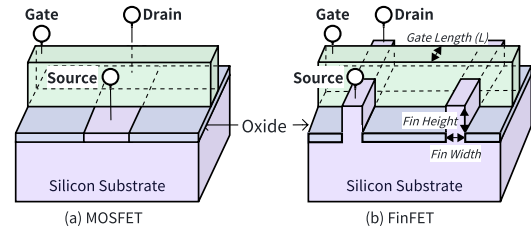


Figure 1. Comparison of planar MOSFET and FinFET structures.

for DC, AC, and transient analyses, where charge-based formulations and continuity constraints underlie reliable predictions [5]. Building an accurate model such as BSIM-CMG [3] requires one equation set that matches device behavior from weak inversion to saturation, with correct small-signal derivatives and charge-derived capacitances. Furthermore, the model must capture key physical responses to terminal voltages [2, 8, 9]: short-channel effects and drain-induced barrier lowering (DIBL), mobility degradation and velocity saturation, quantum confinement and volume inversion in narrow fins, body-bias dependence, self-heating, non-quasi-static (NQS) charge dynamics, and gate leakage plus parasitic overlap/fringe capacitances. Geometry enters via fin count, height, width, and gate length. Across regimes, reciprocity, charge conservation, and smooth transitions ensure that transconductance g_m , output conductance g_{ds} , and capacitances behave consistently. These requirements yield dozens-to-hundreds of coupled parameters with nonlinear interactions and trade-offs [3, 5].

Even with mature compact models, extracting a full parameter set for each geometry and technology node remains challenging. Calibration is high-dimensional and nonconvex, with strong cross-correlations and partial identifiability. Measurements must span wide $V_{gs}/V_{ds}/V_{bs}$ and temperature ranges across geometries, while DC I - V , Q - V , and C - V targets must be met simultaneously under charge conservation and continuity constraints. Consequently, measurement noise, instrument/layout parasitics, and bias-dependent effects (self-heating, NQS) complicate fitting; practical flows still rely on iterative tuning and de-

Table 1. Comparative analysis of mean relative error (MRE) across existing NN-based compact models. The final column reports the average MRE across all characteristics. Architectures: LogicTree (7 levels), ANN (7 layers), MLP (9 layers). Detailed experiments on the encoder-decoder Transformer [7] are presented in Appendix D.

Model	I_{ds}	g_m	g_{ds}	Q_g	Q_b	Q_d	C_{gg}	C_{gd}	C_{gs}	Average
LogicTree	22.21%	32.13%	36.21%	62.31%	14.48%	21.75%	19.42%	23.37%	29.61%	29.05%
ANN [4]	31.78%	24.17%	16.52%	20.30%	22.09%	18.30%	7.74%	14.85%	9.40%	18.35%
MLP [10]	5.18%	16.37%	10.95%	14.80%	9.44%	17.54%	4.94%	12.66%	4.43%	10.70%

vice-specific model cards that do not transfer cleanly across nodes or foundries [3, 5].

Data-driven compact models promise improved accuracy and speed. However, most NN-based approaches still train separate models for I-V, Q-V, and C-V and often maintain distinct models per device type or node; this siloing overlooks joint physical couplings and yields poor transferability across geometries and bias ranges [10]. Scaling up training improves adaptability and accuracy [7], and unified extraction pipelines and binning strategies can enhance calibration and coverage [1, 4]. Nevertheless, a unified generative formulation that models all characteristics jointly, generalizes across devices/technologies, and remains robust to dataset imbalance is still lacking. We benchmark representative NN-based compact models; Table 1 summarizes mean relative error (MRE) across characteristics. In cross-device generalization (train on one device, test on unseen devices), errors rise across characteristics despite reasonable single-device fits, indicating that baselines are insufficiently device-aware and lack robust conditioning on process/design parameters.

To address these challenges, we propose BSIMAR, an autoregressive (AR) compact model that leverages large generative models [6] to overcome limitations of traditional BSIM. Our contributions are threefold:

- **Unified AR modeling and GPU acceleration:** We cast device characterization as next-token prediction. A single AR model jointly generates I-V, Q-V, and C-V with internal consistency, and runs efficiently on modern GPUs to accelerate transistor-level simulation for large AMS circuits.
- **Imbalance-aware weighted loss:** FinFET operating regimes are long-tailed (e.g., subthreshold vs. strong inversion, low vs. high current, extreme bias corners). We use logit-adjusted distribution smoothing (LDS)-weighted mean absolute error (MAE) to rebalance bins and regions, improving tail accuracy and rare-corner fidelity while preserving continuity for circuit simulation.
- **Accuracy via scaling laws:** As an autoregressive foundation model, BSIMAR improves with larger datasets and model sizes, enabling broad pretraining and efficient fine-tuning across devices and technology nodes.

We release all code and trained checkpoints (pretrained foundation and fine-tuned variants) for reproducibility. The

original training data collected from commercial technologies is not released to protect intellectual property.

2. Background

Compact models enable fast, physics-aware device evaluation in circuit simulators, bridging device-level behavior and EDA workflows for DC, AC, and transient analyses. Consequently, they provide consistent currents, charges, and capacitances across biases, geometry, and process.

Developing production-grade compact models demands charge-based, continuous formulations with correct small-signal derivatives, plus calibration across operating regimes and technology nodes; in practice, this pipeline is labor-intensive, node-specific, and acutely sensitive to measurement coverage [5]. BSIM-CMG [3], the industry standard for FinFETs, illustrates both the rigor and the escalating complexity at advanced nodes. Implementations expose large parameter sets (hundreds of parameters) spanning geometry, electrostatics, transport, short-channel effects, self-heating, quantum confinement, variability, and temperature; tens to low hundreds must be calibrated per node and corner, and upkeep scales superlinearly with corner proliferation. Strong cross-coupling induces parameter correlation and non-identifiable subspaces under standard sweeps, leaving extraction ill-conditioned and brittle to coverage, initialization, and heuristics [3, 5].

BSIM-CMG employs a charge-based framework with Ward-Dutton partitioning (charge sharing) to guarantee reciprocity, charge conservation, and smooth small-signal behavior. In compact form, terminal charges satisfy $Q_g + Q_d + Q_s + Q_b = 0$, with $C_{ij} = \frac{\partial Q_i}{\partial V_j}$ and conductances $g_m = \frac{\partial I_{ds}}{\partial V_{gs}}$, $g_{ds} = \frac{\partial I_{ds}}{\partial V_{ds}}$. Geometry and overlap contributions are captured by effective width/perimeter, the partition factor $\alpha(\mathbf{V})$, and overlap charges $Q_{ov,i}$ [3, 5]. While principled, these guarantees incur substantial analytical and numerical complexity.

Calibration typically solves a multi-objective least-squares problem over measured sweeps (I-V,

Q–V, C–V) to recover a parameter vector θ :

$$\min_{\theta} \sum_k \left[w_I \|I_{\text{model}}(\mathbf{V}_k; \theta) - I_{\text{meas}}(\mathbf{V}_k)\|_2^2 + w_Q \|Q_{\text{model}} - Q_{\text{meas}}\|_2^2 + w_C \|C_{\text{model}} - C_{\text{meas}}\|_2^2 \right] \quad (1)$$

subject to continuity and derivative constraints (e.g., smooth g_m , g_{ds}) and charge conservation so that capacitances derived from Q remain consistent across biases. Charge-based formulations enforce these constraints during fitting [5]. However, the high-dimensional, strongly coupled parameterization yields nonconvex objective landscapes; practical extraction depends on regularization, multi-start schedules, and expert priors, and often requires manual triage to avoid non-physical minima.

Furthermore, deep-learning compact models promise to deliver smooth fits, significant speedups, broader coverage via large-scale training, and more scalable extraction [1, 4, 7, 10, 11]. By learning rich, data-driven representations of coupled device behavior, NN-based models can capture complex interactions and reduce hand-tuned parameter burdens. We distill the strengths and limitations into: (i) *Neural compact models (direct modeling)* [10, 11] fit I–V/Q–V/C–V surfaces and accelerate circuit simulation while integrating with BSIM-compatible workflows; however, most target specific characteristics in isolation, weakening explicit relational consistency among I–V, C–V, and Q–V, and therefore require constraints or consistency checks to preserve reciprocity, charge conservation, and smooth derivatives across biases and corners. (ii) *Training scale and extraction/scalability* [1, 4, 7] improve accuracy, adaptability, calibration speed, and coverage—supporting foundation-style models—yet can introduce data/compute demands and distribution-shift sensitivity, and still benefit from careful binning and regularization when coverage is sparse.

In contrast, BSIMAR models all physical responses of FinFETs within a unified autoregressive generative framework over process, design, and voltage sequences, jointly capturing the physical relations among I–V, C–V, and Q–V while maintaining reciprocity, charge conservation, and smooth small-signal behavior. This delivers high accuracy, strong zero-shot generalization, and efficient fine-tuning. Consequently, we compare throughout against the industry-standard BSIM-CMG model [3].

3. Workflow

As illustrated in Figure 2, the BSIMAR model is developed through a structured workflow comprising four main stages:

Table 2. Definitions of source tokens used in dataset generation (process, voltages/temperature, and design), following typical BSIM-CMG conventions.

Category	Parameter	Definition
Process	PHIG	Gate work function; sets flat-band/threshold.
	TOXP	Gate oxide thickness; controls EOT and tunneling.
	U0	Low-field mobility baseline.
	UA	Vertical-field mobility degradation coefficient.
	EU	Mobility field-dependence exponent.
	ETA0	DIBL coefficient; threshold lowering with V_{ds} .
	CIT	Interface-trap capacitance density.
	CGSL	Gate-source overlap/fringe scaling.
V & T	CFS	Source fringing capacitance scaling.
	V_{gs}	Gate-source voltage; controls inversion and transconductance.
	V_{ds}	Drain-source voltage; sets lateral field and region (linear/saturation).
	V_{bs}	Body-source voltage; modulates threshold (body effect).
Design	T	Temperature; affects mobility, velocity saturation, and leakage.
	N_{fin}	Number of fins; scales effective width $W_{\text{eff}} = N_{\text{fin}} P_{\text{eff}}$.
	L	Gate length (L_g); controls short-channel behavior and current.

Parameter Generation We generate a comprehensive dataset by sampling source tokens across three groups: *Process*, *Voltages/Temperature*, and *Design* parameters, as listed in Table 2. To uniformly cover the parameter space with a manageable sample size, we use Latin Hypercube Sampling (LHS). Specifically, we sweep practical ranges for voltages and temperature: $V_{gs} \in [0, 1.0]$ V, $V_{ds} \in [0, 1.0]$ V, $V_{bs} \in [-1, 1]$ V, and $T \in [-25, 125]$ °C; and for design tokens: $N_{\text{fin}} \in \{1, \dots, 24\}$, $L \in [8, 240]$ nm. Each sampled token sequence forms the Source Sequence for the subsequent stages.

SPICE Simulation Using the source sequence, we perform extensive SPICE simulations to generate the corresponding transistor performance characteristics. Outputs are grouped into three categories: I–V (I_{ds} , g_m , g_{ds}), Q–V (Q_g , Q_b , Q_d), and C–V (C_{gg} , C_{gd} , C_{gs}). All definitions appear in Section 2. This collection of simulated data forms the *target sequence*, which is used for loss calculation during model training.

Data Preprocessing First, a *Data Filtering* step removes any sample where at least one of the target tokens (e.g., I_{ds} , g_m , g_{ds}) is exactly zero or falls below a token-wise small-value threshold. This filtering is applied consistently

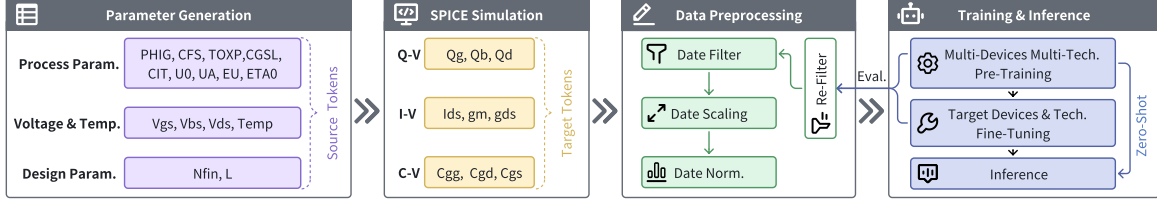


Figure 2. BSIMAR development workflow. A source sequence of diverse process, design, and voltage parameters is used in SPICE to generate a target sequence of I–V, Q–V, and C–V characteristics. After preprocessing, the data drives a two-phase training strategy: a foundation model is created via pretraining and then specialized through efficient fine-tuning. The evaluation loop enables dynamic refinement of the pretraining dataset by monitoring zero-shot evaluation results during training, highlighting the flexibility of the autoregressive framework.

to both the pre-training and fine-tuning datasets, especially critical during fine-tuning; detailed results are provided in [Appendix F](#). Small-valued samples tend to cause training inefficiency and bias optimization toward majority near-zero values; see subsection [Subsection 4.3](#) for our mitigation strategy. Second, all source tokens (including geometric parameters and terminal voltages) are standardized using z-score normalization:

$$\mathbf{x} = \frac{\mathbf{x}' - \mu_{\mathbf{x}'}}{\sigma_{\mathbf{x}'}} \quad (2)$$

where $\mu_{\mathbf{x}'}$ and $\sigma_{\mathbf{x}'}$ are the training-set mean and standard deviation of the original inputs. Third, each target token is independently standardized:

$$y_m = \frac{y'_m - \mu_{y'_m}}{\sigma_{y'_m}}, \quad m = 1, \dots, M \quad (3)$$

where M is the number of target tokens, and $\mu_{y'_m}$, $\sigma_{y'_m}$ are computed per token on training data only. This preserves the distinct scale and distribution of each physical quantity while enabling stable autoregressive learning.

Training & Inference Our training strategy is a two-phase process designed for maximum accuracy and adaptability. First, we conduct *Multi-Device Multi-Technology Pre-training* on a vast and diverse dataset, allowing BSIMAR to learn a robust, generalized foundation of transistor physics. While the pre-trained model is powerful, we subsequently perform *Target Device & Technology Fine-tuning*. This fine-tuning capability is particularly valuable when available data is severely restricted ([Figure 7](#)). For instance, when developing novel device prototypes or working with limited foundry-provided data, generating a large dataset is often impractical or prohibitively expensive. In these scenarios, BSIMAR can be fine-tuned with only a small number of data points to create a highly accurate, specialized model. BSIMAR also supports zero-shot inference, with performance comparisons listed in [Table 3](#), showcasing strong generalization.

4. BSIMAR

We now detail the architecture, training objective, and cross-device adaptation for BSIMAR, using unified notation: inputs $\mathbf{x}$ (source tokens), outputs $\mathbf{y}$ (target tokens), predictions $\hat{\mathbf{y}}$, with M target tokens and batch set $\mathcal{B}$.

4.1. Model Architecture

BSIMAR uses a decoder-only autoregressive (AR) Transformer (see [Figure 3](#)). Masked multi-head self-attention with a feed-forward network (FFN) learns dependencies among tokens. Conditioned on source tokens (inputs) $\mathbf{x}$, the model generates target tokens (outputs) $\mathbf{y}$ by factorizing the joint probability:

$$p(y_1, \dots, y_M | \mathbf{x}) = \prod_{i=1}^M p(y_i | \mathbf{x}, y_1, \dots, y_{i-1}) \quad (4)$$

This formulation treats each target token as a continuous scalar and enforces coherence across I–V, Q–V, and C–V predictions.

4.2. Model Training

During training, the model processes a sequence composed of source tokens (inputs) and target tokens (outputs) under a causal attention mask. Because the mask enforces autoregressive dependencies, the order of output tokens matters. We obtain the best results when the order follows BSIM-CMG’s physical derivative/integral chain: first Q–V (charges), then I–V (currents), and finally C–V (capacitances). Each token is represented by concatenating its scalar value with a learned position embedding that identifies the token type and index; this resolves permutation invariance in self-attention and supports the physically-motivated ordering. The model is optimized with the LDS-weighted MAE objective described below.

4.3. Smoothed MAE Loss

The histograms in [Figure 4](#) show strong skew and imbalance across Q–V, I–V, and C–V: data cluster in common

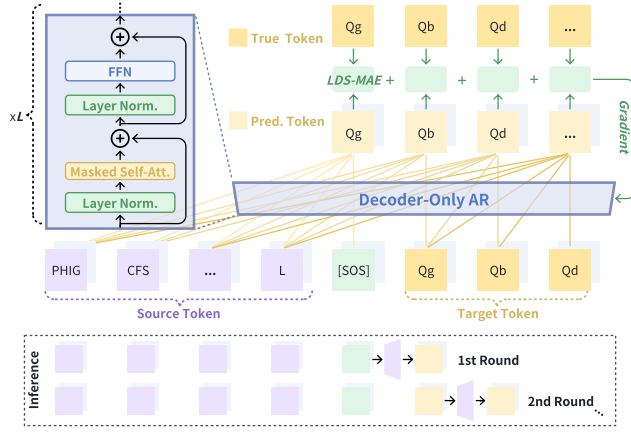


Figure 3. BSIMAR architecture and training/inference. A decoder-only AR Transformer (middle) with masked self-attention and an FFN. During training, the sequence concatenates source tokens and target tokens. During inference, the model generates one target token per step and appends it to condition the next prediction.

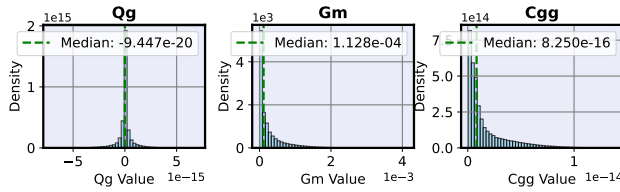


Figure 4. Long-tailed label distribution of different Q-V, I-V, C-V characteristics of the collected training set.

operating regions, with long tails covering rare regimes. This arises from two coupled factors consistent with § 2 and § ??: (i) input P-V-T sampling and design choices concentrate around typical biases (V_{gs} , V_{ds} , V_{bs}), geometries, and process corners, yielding dense head regions while rare corner combinations receive fewer measurements; and (ii) FinFET device physics are highly nonlinear—subthreshold I-V is exponential in V_{gs} , near-threshold exhibits sharp curvature, C-V reflects depletion/accumulation and overlap contributions, Q-V integrates charge partitioning, and temperature/self-heating/NQS effects stretch tails—so labels spread unevenly across regimes [3, 5]. Under these long-tailed densities, uniform losses bias training toward head regions and underfit tails.

To counter this, we adopt Label Distribution Smoothing (LDS) [12] and optimize a weighted MAE. In brief, LDS constructs per-sample weights proportional to the inverse of a kernel-smoothed estimate of the marginal label density for each target token, thereby flattening risk across bins. Concretely, for each target index $m \in \{1, \dots, M\}$ we discretize y_m into n_{bins} uniform bins to obtain counts $c_m[j]$, then smooth them by convolving with a short window $k[t]$

(length k_s , bandwidth σ):

$$\tilde{c}_m[j] = \sum_{t=-h}^h k[t] c_m[j-t], \quad h = \frac{k_s-1}{2}, \quad (5)$$

where $k[t]$ is Gaussian with $k_s=5$ and $\sigma=0.8$. For a sample $i \in \mathcal{B}$ in bin j of token y_m , we set a single clipped, mean-normalized weight:

$$w_m^{(i)} = \text{clip} \left(\frac{1}{\tilde{c}_m[j] + \varepsilon} / \frac{1}{|\mathcal{B}|} \sum_{i' \in \mathcal{B}} \frac{1}{\tilde{c}_m[j(i')] + \varepsilon}, w_{\min}, w_{\max} \right) \quad (6)$$

and optimize the weighted MAE

$$\mathcal{L}_{\text{LDS-MAE}} = \frac{1}{M} \sum_{m=1}^M \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} w_m^{(i)} |\hat{y}_m^{(i)} - y_m^{(i)}|. \quad (7)$$

Intuition and comparison. The weight $w \propto 1/\tilde{p}(y)$ acts like importance weighting against the (smoothed) label density, so Eq. (7) approximates a density-flattened risk that allocates capacity to rare operating regimes. In contrast, MSE compounds large residuals with high head-region frequency, yielding biased gradients and poor tail calibration. MAE provides bounded per-sample gradients; combined with LDS, it improves tail accuracy and stabilizes cross-device optimization (Table 5). We observe Gaussian windows ($k_s=5$, $\sigma=2.0$) to be the most stable among triangular/Laplace alternatives, and per-dimension standardization (3) further keeps gradients well-conditioned. Additional experiments on advanced loss functions are presented in the supplementary material Appendix E.

4.4. Few-Shot Cross-Device/Cross-Technology Modeling

BSIMAR models the physical response of FinFET devices—not merely a surrogate of given devices—by learning the structured dependence of multi-physics outputs (I_{ds} , Q_g , C_{gg} , g_m , etc.) on bias and technology factors. Notably, BSIMAR is also applicable to planar MOSFETs; experiments on MOSFET devices are provided in Appendix D. It enables few-shot adaptation across device flavors within a node (e.g., 16 nm nch_svt, nch_lvt, pch_hvt) and across technology generations. With only a small set of I-V/C-V sweeps from the target (typically fewer than 2K bias points), BSIMAR recovers unmeasured operating regions and coupled outputs without full compact-model re-extraction. Comparative results of few-shot experiments incorporating Advanced Loss are presented in Appendix E of the supplementary material.

5. Results

Results overview. We organize the findings into scaling behavior, prediction accuracy before and after fine-tuning,

data-size sensitivity, and cross-technology/device trends. Unless noted, “zero-shot” denotes the pretrained general model without device-specific fine-tuning, and “+ft” denotes specialized fine-tuning. Based on the BSIMAR framework, comprehensive circuit-level validation and detailed experimental results are provided in Appendix C of the supplementary material.

5.1. Hyperparameters

All models were trained with a batch size of 1024 and optimized using AdamW with a pretraining learning rate of 8×10^{-4} and weight decay of 1×10^{-4} . Early stopping was applied with a patience of 10 epochs during pretraining, using a minimum improvement threshold of 1×10^{-5} . Architecturally, the models share a hidden dimension of 256, feedforward dimension of 1024, 8 attention heads, and a dropout rate of 0.2, differing only in depth: ModelS(Model Size=3.16M) has 4 layers, ModelM(Model Size=3.95M) has 5, and ModelL(Model Size=4.47M) has 6. Detailed hyperparameters for the model’s pre-training and fine-tuning stages are presented in the table in Appendix B

5.2. Scaling Law of BSIMAR

Figure 5 illustrates BSIMAR’s favorable scaling behavior with respect to data and model size. Increasing the training set from 1.44M to 18.4M samples reduces the average Mean Relative Error (MRE) monotonically from 29.68% to 12.98%, demonstrating strong data efficiency without saturation. With the dataset size fixed at 18.4M, scaling the model capacity from Small (S) to Medium (M) to Large (L) consistently improves performance, reducing MRE from 23.28% to 12.98% with no observed slowdown or instability. These results indicate that BSIMAR remains under-parameterized relative to task complexity and benefits reliably from joint scaling of data and model size.

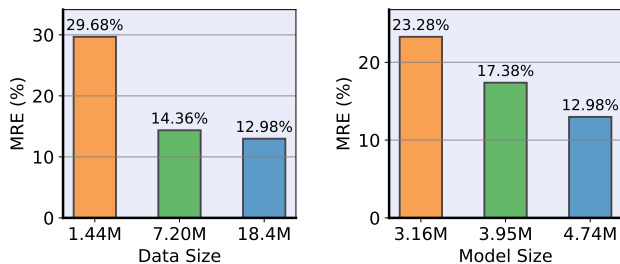


Figure 5. BSIMAR zero-shot accuracy as model/data scale increases. Avg. MRE decreases monotonically with both data size and model capacity; no slowdown of MRE with increasing model size is observed in the tested regime.

5.3. Prediction Accuracy: Zero-Shot vs. Fine-Tuning

The waveform comparison in Figure 6 demonstrates high-fidelity emulation of FinFET device physics by the BSIMAR model across multiple electrostatic and dynamic parameters. The close alignment between predicted (blue) and real (orange) traces indicates strong agreement in both qualitative behavior and quantitative accuracy, particularly for Q_g , I_{ds} , C_{gg} , C_{gd} , and C_{gs} , where deviations are almost all within 10% of the signal magnitude.

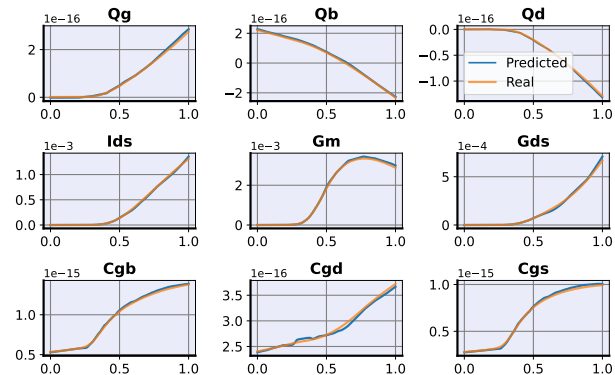


Figure 6. BSIMAR predictions versus BSIM-CMG (7nm nch_svt) ground truth across representative I-V, Q-V, and C-V characteristics. Curves show close alignment in both magnitude and trend, illustrating zero-shot fidelity and the effect of fine-tuning.

Table 3 (MRE, %) and Table 4 (R^2) jointly evaluate BSIMAR across three technologies (7/12/16 nm), four device flavors (nch_svt, nch_lvt, pch_svt, pch_lvt), and nine targets covering I-V, Q-V, and C-V. Rows reflect the same methodology used earlier (zero-shot vs. fine-tuned, using LDS-weighted MAE), ensuring consistency in how accuracy is assessed. Fine-tuning reduces the average MRE in every device family across 7/12/16 nm, typically by 40%–70% relative. Gains are most pronounced for transport and charge targets (e.g., I_{ds} , g_m , Q_g , Q_b), while occasional small-signal/feedback regressions (e.g., g_{ds} , C_{gd} , C_{gg}) arise in long-tailed, low-signal regions. Both svt and lvt benefit; 16 nm exhibits the largest relative drops, followed by 12 nm and 7 nm. Zero-shot averages are already single-digit in several cases; fine-tuning consolidates those into consistently low errors across families.

Fine-tuning consistently reduces the average Mean Relative Error (MRE) by 50–70%, achieving 2–4% error rates across multiple technology nodes and both svt/lvt device variants. The most substantial improvements occur where zero-shot performance was weakest, notably at the 16 nm node, effectively correcting technology-specific modeling biases. Accuracy gains are especially pronounced for static characteristics: transport behavior (I - V curves) and elec-

Table 3. Comparison of BSIMAR’s Zero-Shot vs. Fine-Tuning performance, measured by Mean Relative Error (MRE, %), across 4 FinFET devices and 3 technologies. Green/Red indicates improvement/degradation.

Tech. Device		I-V			Q-V			C-V			Average
		I_{ds}	g_m	g_{ds}	Q_g	Q_b	Q_d	C_{gg}	C_{gd}	C_{gs}	
7nm	nch_svt	2.22%	6.00%	10.47%	7.42%	30.42%	4.90%	1.55%	3.57%	6.47%	8.10%
	nch_svt+ft	1.32% -0.90%	1.61% -4.39%	12.81% +2.34%	3.51% -3.91%	5.02% -25.40%	4.15% -0.65%	1.79% +0.24%	1.89% -1.68%	2.49% -3.98%	3.84% -4.26%
	nch_lvt	10.04%	16.20%	6.64%	11.53%	9.99%	9.61%	5.24%	1.57%	7.11%	6.41%
	nch_lvt+ft	1.77% -8.27%	1.57% -14.63%	6.76% +0.12%	7.57% -3.96%	5.48% -4.51%	8.74% -0.87%	1.19% -4.05%	2.83% +1.26%	2.73% -4.38%	4.29% -2.12%
	pch_svt	10.19%	5.39%	10.64%	6.12%	3.63%	7.33%	4.16%	2.62%	7.91%	6.44%
	pch_svt+ft	6.95% -3.24%	10.83% +5.44%	3.75% -6.89%	8.30% +2.18%	3.76% +0.13%	7.16% -0.17%	4.10% -0.06%	4.14% +1.52%	7.58% -0.33%	6.29% -0.15%
	pch_lvt	9.77%	14.96%	20.53%	10.69%	6.75%	13.88%	2.04%	2.08%	4.97%	9.52%
pch_lvt+ft	6.62% -3.15%	14.58% -0.38%	23.24% +2.71%	4.31% -6.38%	11.67% +4.92%	7.33% -6.55%	1.95% -0.09%	2.54% +0.46%	4.93% -0.04%	8.58% -0.94%	
12nm	nch_svt	1.78%	5.05%	12.89%	8.72%	18.43%	7.06%	1.98%	2.27%	3.37%	6.84%
	nch_svt+ft	2.89% +1.11%	3.13% -1.92%	12.80% -0.09%	3.26% -5.46%	1.66% -16.77%	5.25% -1.81%	2.69% +0.71%	1.11% -1.16%	4.08% +0.71%	4.10% -2.74%
	nch_lvt	0.45%	7.71%	7.65%	7.08%	12.43%	7.31%	1.68%	0.85%	3.07%	5.36%
	nch_lvt+ft	2.04% +1.59%	2.87% -4.84%	7.65% +0.00%	1.63% -5.45%	5.73% -6.70%	6.49% -0.82%	0.92% -0.76%	1.12% +0.27%	2.16% -0.91%	3.54% -1.82%
	pch_svt	3.96%	10.16%	8.83%	17.44%	6.79%	6.37%	0.83%	2.06%	2.58%	6.56%
	pch_svt+ft	5.49% +1.53%	4.75% -5.41%	6.88% -1.95%	3.63% -13.81%	3.68% -3.11%	2.52% -3.85%	0.81% -0.02%	0.60% -1.46%	1.61% -0.97%	3.33% -3.23%
	pch_lvt	3.09%	6.12%	9.86%	20.47%	10.42%	11.87%	1.39%	1.44%	3.46%	7.57%
pch_lvt+ft	3.86% +0.77%	4.28% -1.84%	5.09% -4.77%	3.05% -17.42%	2.96% -7.46%	7.49% -4.38%	0.89% -0.50%	0.85% -0.59%	1.17% -2.29%	3.29% -4.28%	
16nm	nch_svt	11.88%	18.27%	14.13%	4.04%	4.21%	15.41%	4.17%	2.71%	4.44%	8.81%
	nch_svt+ft	2.15% -9.73%	1.59% -16.68%	10.48% -3.65%	2.82% -1.22%	7.35% +3.14%	7.08% -8.33%	1.98% -2.19%	0.90% -1.81%	1.81% -2.63%	4.02% -4.79%
	nch_lvt	7.82%	10.83%	14.61%	2.10%	8.46%	15.28%	3.27%	1.87%	2.59%	7.42%
	nch_lvt+ft	2.88% -4.94%	2.06% -8.77%	11.69% -2.92%	0.91% -1.19%	2.40% -6.06%	8.87% -6.41%	1.51% -1.76%	1.01% -0.86%	1.94% -0.65%	3.70% -3.72%
	pch_svt	17.46%	8.57%	10.52%	3.64%	3.46%	28.75%	2.22%	2.30%	2.53%	8.83%
	pch_svt+ft	2.62% -14.84%	3.12% -5.45%	3.28% -7.24%	6.19% +2.55%	3.84% +0.38%	1.83% -26.92%	0.90% -1.32%	0.72% -1.58%	0.98% -1.55%	2.61% -6.22%
	pch_lvt	15.05%	9.38%	11.54%	8.56%	6.54%	28.97%	1.81%	1.78%	1.37%	9.44%
pch_lvt+ft	3.60% -11.45%	4.39% -4.99%	5.79% -5.75%	4.90% -3.66%	1.54% -5.00%	3.79% -25.18%	0.82% -0.99%	1.60% -0.18%	2.08% +0.71%	3.17% -6.27%	

Table 4. Comparison of BSIMAR’s Zero-Shot vs. Fine-Tuning performance, measured by the Coefficient of Determination (R^2), across 4 FinFET devices and 3 technologies. R^2 indicates a perfect correlation between predictions and the true data.

Tech. Device		I-V			Q-V			C-V			
		I_{ds}	g_m	g_{ds}	Q_g	Q_b	Q_d	C_{gg}	C_{gd}	C_{gs}	Average
7nm	nch_svt	0.998	0.992	0.928	0.995	0.938	0.993	0.997	0.662	0.973	0.942
	nch_svt+ft	0.991 ^{-0.007}	0.990 ^{-0.002}	0.937 ^{+0.009}	0.990 ^{-0.005}	0.988 ^{+0.050}	0.996 ^{+0.004}	0.956 ^{-0.041}	0.956 ^{+0.293}	0.997 ^{+0.024}	0.982 ^{+0.041}
	nch_lvt	0.967	0.913	0.962	0.957	0.992	0.954	0.946	0.911	0.932	0.948
	nch_lvt+ft	0.998 ^{+0.032}	0.999 ^{+0.085}	0.923 ^{-0.039}	1.000 ^{+0.043}	0.997 ^{+0.005}	0.999 ^{+0.045}	0.999 ^{+0.054}	0.878 ^{-0.033}	0.998 ^{+0.066}	0.977 ^{+0.029}
	pch_svt	0.948	0.979	0.982	0.999	0.989	0.996	0.953	0.856	0.943	0.960
	pch_svt+ft	0.979 ^{+0.032}	0.935 ^{-0.044}	0.981 ^{-0.001}	0.987 ^{-0.012}	0.997 ^{+0.008}	0.971 ^{-0.025}	0.960 ^{+0.007}	0.739 ^{-0.117}	0.952 ^{+0.010}	0.945 ^{-0.016}
	pch_lvt	0.941	0.755	0.541	0.996	0.993	0.976	0.988	0.916	0.972	0.898
	pch_lvt+ft	0.983 ^{+0.041}	0.807 ^{+0.052}	0.204 ^{-0.337}	0.995 ^{-0.001}	0.993 ^{-0.000}	0.996 ^{+0.020}	0.991 ^{+0.003}	0.869 ^{-0.047}	0.980 ^{+0.009}	0.869 ^{-0.029}
12nm	nch_svt	0.997	0.988	0.795	0.986	0.849	0.996	0.989	-0.511	0.991	0.787
	nch_svt+ft	0.996 ^{-0.001}	0.996 ^{+0.007}	0.844 ^{+0.049}	1.000 ^{+0.013}	0.998 ^{+0.149}	0.997 ^{+0.001}	0.976 ^{-0.013}	0.710 ^{+1.221}	0.987 ^{-0.004}	0.945 ^{+0.158}
	nch_lvt	0.995	0.981	0.953	0.973	0.952	0.996	0.997	0.694	0.993	0.948
	nch_lvt+ft	0.998 ^{+0.003}	0.994 ^{+0.012}	0.946 ^{-0.007}	1.000 ^{+0.027}	0.997 ^{+0.045}	0.995 ^{-0.001}	0.997 ^{+0.000}	0.863 ^{+0.168}	0.995 ^{+0.002}	0.976 ^{+0.028}
	pch_svt	0.995	0.981	0.953	0.973	0.952	0.996	0.997	0.694	0.993	0.948
	pch_svt+ft	0.988 ^{-0.007}	0.992 ^{+0.011}	0.970 ^{+0.017}	0.999 ^{+0.026}	0.985 ^{+0.033}	0.997 ^{+0.001}	0.998 ^{+0.001}	0.973 ^{+0.278}	0.997 ^{+0.005}	0.989 ^{+0.041}
	pch_lvt	0.988	0.992	0.896	0.953	0.964	0.984	0.995	0.876	0.983	0.959
	pch_lvt+ft	0.983 ^{-0.005}	0.993 ^{+0.002}	0.965 ^{+0.070}	0.999 ^{+0.046}	0.995 ^{+0.031}	0.997 ^{+0.013}	0.997 ^{+0.002}	0.950 ^{+0.074}	0.998 ^{+0.015}	0.986 ^{+0.027}
16nm	nch_svt	0.966	0.910	0.816	0.992	0.998	0.941	0.940	-0.776	0.968	0.750
	nch_svt+ft	0.999 ^{+0.033}	0.999 ^{+0.090}	0.879 ^{+0.063}	0.998 ^{-0.004}	0.996 ^{-0.002}	0.994 ^{+0.053}	0.987 ^{+0.047}	0.766 ^{+1.542}	0.997 ^{+0.029}	0.957 ^{+0.207}
	nch_lvt	0.976	0.957	0.910	1.000	0.994	0.976	0.952	0.455	0.987	0.913
	nch_lvt+ft	0.994 ^{+0.018}	0.998 ^{+0.040}	0.891 ^{-0.019}	1.000 ^{+0.000}	0.999 ^{+0.005}	0.991 ^{+0.015}	0.991 ^{+0.039}	0.839 ^{+0.384}	0.996 ^{+0.008}	0.966 ^{+0.053}
	pch_svt	0.874	0.965	0.997	0.999	0.991	0.941	0.979	0.393	0.991	0.903
	pch_svt+ft	0.996 ^{+0.122}	0.998 ^{+0.033}	0.993 ^{-0.004}	0.998 ^{-0.000}	0.969 ^{-0.022}	0.998 ^{+0.057}	0.997 ^{+0.018}	0.940 ^{+0.547}	0.999 ^{+0.008}	0.988 ^{+0.084}
	pch_lvt	0.921	0.945	0.982	0.996	0.979	0.945	0.979	0.753	0.997	0.944
	pch_lvt+ft	0.990 ^{+0.069}	0.996 ^{+0.051}	0.980 ^{-0.002}	0.999 ^{+0.003}	0.997 ^{+0.018}	0.996 ^{+0.051}	0.998 ^{+0.018}	0.809 ^{+0.056}	0.994 ^{-0.003}	0.973 ^{+0.029}

416 trostatics (Q - V relationships) show marked enhancement.
 417 Dynamic capacitance metrics also improve broadly, with
 418 clear gains in C_{gg} and C_{gs} . The largest corrections are ob-
 419 served in feedback capacitance C_{gd} at 12/16 nm, although
 420 minor regressions remain in 7 nm p-type small-signal con-

ductance (g_{ds}) and feedback regions. Concurrently, pre-
 dictive correlations recover significantly: R^2 values ap-
 proach unity across nearly all device families. In particu-
 lar, C_{gd} correlations at 12/16 nm—previously weak or neg-
 ative—now exhibit strong positive alignment, with only

421
422
423
424
425

Table 5. LDS-MAE loss ablation at the **pre-training** stage: Mean Relative Error (MRE, %) across BSIMAR predictions for n -channel and p -channel SVT devices at 7nm. Results compare training objectives: MSE, MAE, and MAE+LDS.

Tech. & Device	Loss	I_{ds}	g_m	g_{ds}	Q_g	Q_b	Q_d	C_{gg}	C_{gd}	C_{gs}	Average
7nm nch_svt	MSE	2.35%	11.22%	25.44%	23.10%	8.78%	20.73%	6.02%	7.92%	10.34%	12.88%
	MAE	4.16%	6.70%	41.67%	13.54%	8.98%	14.35%	3.24%	2.24%	8.24%	11.46%
	MAE+LDS	2.22%	6.00%	10.47%	7.42%	30.42%	4.80%	1.55%	3.57%	6.47%	8.10%
7nm pch_svt	MSE	9.48%	8.84%	14.81%	27.55%	38.02%	24.59%	7.43%	3.64%	15.49%	16.65%
	MAE	1.71%	11.15%	34.79%	6.75%	15.70%	20.07%	4.15%	4.85%	8.38%	11.95%
	MAE+LDS	10.19%	5.39%	10.64%	6.12%	3.63%	7.33%	4.16%	2.62%	7.91%	6.44%

a few isolated outliers persisting in 7 nm p-type g_{ds} , underscoring the robustness and generalizability of the fine-tuning approach.

Generalized insights: To summarize across technologies and operating regimes: (a) **Cross-regime consistency:** The autoregressive architecture leverages cross-family couplings to jointly enhance transport, electrostatics, and dynamics; gains are largest at nodes with stronger nonidealities (16 nm). (b) **Technology-specific residuals:** Nevertheless, remaining errors cluster in 7 nm p-type feedback/small-signal regimes, driven by long-tailed distributions and rare operating corners. (c) **Deviation origin and mitigation:** In light of (a) and (b), regressions align with class imbalance and small-value filtering; consequently, LDS-weighted MAE mitigates head-dominance, while per-target weighting and curriculum sampling preserve derivative fidelity without degrading average MRE.

Relative to prior NN-based compact models (Table 1), BSIMAR achieves 2–4% fine-tuned MRE—delivering a $2\times$ – $4\times$ accuracy advantage over baselines (10–30% MRE)—while zero-shot already matches or exceeds stronger counterparts in multiple devices. Near-unity R^2 after fine-tuning resolves prior cross-device generalization gaps, corroborating the scaling-law and specialization paradigm.

5.4. Data Size for Fine-Tuning

Figure 7 summarizes MRE versus fine-tuning set size (10^3 , 2×10^3 , 4×10^3 , 8×10^3) across nine FinFET targets grouped into current-voltage (I-V), charge-voltage (Q-V), and capacitance-voltage (C-V). All curves decrease steadily, with the steepest relative reductions from 10^3 to 2×10^3 and from 2×10^3 to 4×10^3 . Transport and charge quantities (e.g., I_{ds} , g_m , Q_g , Q_b) reach single-digit MRE quickly, while feedback and small-signal terms (g_{ds} , C_{gd} , C_{gs} , C_{gg}) improve more gradually and show mild diminishing returns beyond 4×10^3 . Overall, BSIMAR is data-efficient: at 2×10^3 samples it achieves a large fraction of the total improvement, and moving to 4×10^3 – 8×10^3 yields additional reductions without instability, consistent with long-tailed label density and derivative sensitivity.

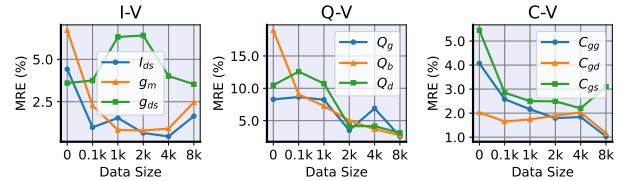


Figure 7. The prediction MRE consistently decreases across a suite of device characteristic metrics as we increase the number of fine-tuning data samples up to 8K.

5.5. Ablation Study for LDS-MAE

Table 5 (pre-training) compare three objectives—MSE, MAE, and MAE+LDS—on 7 nm svt devices (n/p), reporting MRE across I-V, Q-V, and C-V with an overall average. The setup aligns with our methodology and isolates how imbalance-aware training reshapes accuracy across long-tailed regimes and small-value regions.

Across pre-training, MAE+LDS consistently yields the lowest average MRE for both n - and p -channel devices, with gains concentrated in tail-sensitive and derivative quantities (e.g., g_{ds} , Q_g , Q_b); pure MAE occasionally performs better for select small-signal capacitances. A minor exception for n -type Q_b under pre-training is resolved after fine-tuning.

Fine-tuning further reduces MRE across device flavors, amplifying improvements in derivatives and distribution tails while stabilizing electrostatics and capacitance predictions. Remaining outliers are confined to a few small-signal/feedback entries, with overall trends robust across targets.

Generalized conclusions. Taken together, the results suggest three practical takeaways: (a) Consistency across regimes: Imbalance-aware MAE+LDS jointly enhances transport, electrostatics, and dynamics; gains are most pronounced in long-tailed regions and nonideal operating corners. (b) Device variation: However, p -type benefits more during pre-training, whereas n -type derivatives improve most during fine-tuning; residual errors persist in small-signal/feedback regimes due to extreme label sparsity and rare corner combinations. (c) Deviation analysis and mitigation: Consequently, exceptions correlate with class imbalance and small-value filtering; combining LDS

with per-target weighting and curriculum sampling preserves localized derivative fidelity without degrading average MRE.

6. Conclusions & Future Work

We presented BSIMAR, a decoder-only autoregressive compact model that scales with data and model size, achieves strong zero-shot fidelity, and realizes further gains with modest fine-tuning budgets.

Limitations. Evaluation relies on BSIM-CMG simulations (no measured-silicon validation), covers rare corners and small-signal/feedback regimes incompletely, focuses on FinFET `svt/lvt` and limited nodes, lacks optimized inference throughput and mature SPICE/EDA integration, and does not enforce physics priors/bounds—extrapolation remains unverified.

Future work. Unify parameter extraction via inverse inference; broaden scaling/generalization across nodes and devices with imbalance-aware training and per-target weighting; accelerate inference (distillation, Mixture of Experts) and harden SPICE/EDA integration for practical deployment.

References

- [1] Jinyoung Choi, Hyunjoon Jeong, Sangmin Woo, Hyungmin Cho, Yohan Kim, Jeong-Taek Kong, and Soyoung Kim. Enhancement and expansion of the neural network-based compact model using a binning method. *IEEE Journal of the Electron Devices Society*, 12:65–73, 2024. 2, 3
- [2] Jean-Pierre Colinge. *FinFETs and Other Multi-Gate Transistors*. Springer, New York, 2008. 1
- [3] Juan P Duarte, Sourabh Khandelwal, Aditya Medury, Chenming Hu, Pragma Kushwaha, Harshit Agarwal, Avirup Dasgupta, and Yogesh S Chauhan. Bsim-cmg: Standard finfet compact model for advanced circuit design. In *ESSCIRC Conference 2015-41st European Solid-State Circuits Conference (ESSCIRC)*, pages 196–201. IEEE, 2015. 1, 2, 3, 5
- [4] Hyunjoon Jeong, Sangmin Woo, Jinyoung Choi, Hyungmin Cho, Yohan Kim, Jeong-Taek Kong, and Soyoung Kim. Fast and expandable ann-based compact model and parameter extraction for emerging transistors. *IEEE Journal of the Electron Devices Society*, 11:153–160, 2023. 2, 3
- [5] D Lu. Ph. d. dissertation: Compact models for future generation cmos. *Electrical Engineering and Computer Sciences University of California at Berkeley*, 2011. 1, 2, 3, 5
- [6] Ziqi Pang, Tianyuan Zhang, Fujun Luan, Yunze Man, Hao Tan, Kai Zhang, William T Freeman, and Yu-Xiong Wang. Randar: Decoder-only autoregressive visual generation in random orders. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 45–55, 2025. 2
- [7] Chanwoo Park, Seungjun Lee, Junghwan Park, Kyungjin Rim, Jihun Park, Seonggook Cho, Jongwook Jeon, and Hyunbo Cho. Large-scale training in neural compact models for accurate and adaptable mosfet simulation. *IEEE Journal of the Electron Devices Society*, 2024. 2, 3, 1
- [8] S. M. Sze and Kwok K. Ng. *Physics of Semiconductor Devices*. Wiley, Hoboken, NJ, 3rd edition, 2007. 1
- [9] Yannis Tsividis. *Operation and Modeling of the MOS Transistor*. Oxford University Press, New York, 2nd edition, 1999. 1
- [10] Chien-Ting Tung and Chenming Hu. Neural network-based bsim transistor model framework: Currents, charges, variability, and circuit simulation. *IEEE Transactions on Electron Devices*, 70(4):2157–2160, 2023. 2, 3
- [11] Chien-Ting Tung, Ming-Yen Kao, and Chenming Hu. Neural network-based compact modeling with high accuracy and potential model speed. *IEEE Transactions on Electron Devices*, 69:6476–6479, 2022. 3
- [12] Yuzhe Yang, Kaiwen Zha, Yingcong Chen, Hao Wang, and Dina Katabi. Delving into deep imbalanced regression. In *International conference on machine learning*, pages 11842–11851. PMLR, 2021. 5

BSIMAR: An Autoregressive Compact Model for FinFETs

Supplementary Material

This supplementary is organized to mirror the flow of results. Section A introduces the baseline compact models and datasets. Section A.1 reports accuracy across technologies with error breakdowns, and Section A.2 contrasts scaling laws for encoder-decoder and decoder-only architectures. Section B details BSIMAR training protocols, hyperparameters, and preprocessing. Section C describes circuit-simulation integration, including DC and transition analyses. Section D presents an extended study on planar MOSFETs. Section E compares loss functions (MAE, MSE, and LDS-MAE). Finally, we outline our data-filtering strategy and its rationale.

A. Baseline Compact Models

A.1. Accuracy

Building on the baseline in ?? (Table 1), Table 6 and Table 7 report fine-tuned results for four neural architectures—LogicTree (7 levels), ANN (7 layers), MLP (9 layers), and the encoder-decoder transformer [7] (E-D, 6 layers)—across small-signal and charge families. Comparing the introductory baseline with the fine-tuned outcomes shows a consistent trend: average MRE declines for all models (LogicTree 29.05% $\rightarrow$ 26.07%, ANN 18.35% $\rightarrow$ 13.37%, MLP 10.70% $\rightarrow$ 9.62%), with the E-D transformer achieving the lowest average MRE after fine-tuning (8.65%).

Across I - V , Q - V , and C - V , errors vary by quantity (Table 6). E-D shows particularly low MRE for the derivative-sensitive g_{ds} (3.17%), and capacitance errors for C_{gd} are competitive across models (ANN 1.88%, MLP 4.00%, E-D 4.28%). Overall, E-D attains the lowest average MRE, with MLP and ANN close behind.

Correlation results in Table 7 reveal architectural trade-offs: ANN attains the highest average R^2 ($\approx$ 0.864), followed by MLP ($\approx$ 0.850), whereas E-D, despite the lowest MRE, shows weaker correlation on specific small-signal metrics (e.g., C_{gd} with a slightly negative value). Lower relative error can coincide with reduced pointwise correlation in sharply curved regimes, especially for capacitances.

Key observations: (i) among the fine-tuned architectures, E-D achieves the lowest average MRE; (ii) ANN and MLP exhibit stronger average R^2 , indicating better global alignment on small-signal responses; (iii) low or negative R^2 in C_{gd} (notably for E-D) underscores the importance of enforcing cross-characteristic consistency (I - V , Q - V , C - V).

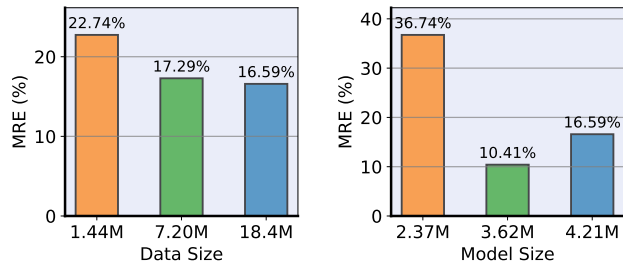


Figure 8. Encoder-decoder transformer scaling. In our reconstruction of [7], average MRE decreases monotonically with data size but shows weak or non-monotonic improvements with increased model capacity.

A.2. Encoder-Decoder vs. Decoder-Only

To differentiate the two model architectures, we compare the encoder-decoder transformer (Figure 8) with the decoder-only BSIMAR (Figure 5). Both models improve with more data. The encoder-decoder is trained in the same way as BSIMAR, to predict all target tokens in one pass. Nevertheless, capacity scaling is weak or non-monotonic. We trace these effects to two root causes: (i) inductive bias, with cross-attention preferentially modeling dense head regions while under-weighting rare tails, limiting effective capacity utilization; (ii) optimization sensitivity, as larger encoder-decoder variants require re-tuned learning rates, weight decay, and normalization to avoid gradient-noise plateaus.

BSIMAR scales cleanly with both data and capacity (Figure 5) because a unified decoder and lightweight cross-family consistency constraints couple currents, charges, and capacitances more strongly than sequence-level coupling alone, stabilizing derivatives and aligning heads with tails. Together with curvature-aware smoothing (e.g., LDS-MAE) and shared parameterization, capacity is spent on rare regimes rather than re-fitting the head, yielding lower average MRE in zero-shot and adapted settings at modest memory/compute. This physically guided inductive bias explains the stronger tail coverage and more efficient capacity utilization compared to the encoder-decoder and prior work [7].

B. BSIMAR Hyperparameters

We summarize the hyperparameter choices used across all BSIMAR experiments and explain the rationale behind each setting. To isolate the effect of scaling while keeping memory stable, we vary model *depth* across three capacity points (Model S/M/L), while holding width and atten-

Table 6. Fine-tuning performance of mean relative error (MRE, %) across existing NN-based compact models. Architectures: ANN (7 layers), MLP (9 layers), E-D (6 layers)

Model	I_{ds}	g_m	g_{ds}	Q_g	Q_b	Q_d	C_{gg}	C_{gd}	C_{gs}	Average
ANN	21.58%	21.78%	11.63%	30.26%	16.88%	6.13%	4.20%	1.88%	6.00%	13.37%
MLP	12.56%	15.66%	9.33%	20.41%	4.76%	11.62%	3.39%	4.00%	4.85%	9.62%
E-D	10.39%	12.12%	3.17%	16.02%	5.41%	18.11%	4.07%	4.28%	4.26%	8.65%

Table 7. Fine-tuning performance of Determination (R^2) across existing NN-based compact models.

Model	I_{ds}	g_m	g_{ds}	Q_g	Q_b	Q_d	C_{gg}	C_{gd}	C_{gs}	Average
ANN	0.841	0.951	0.984	0.845	0.668	0.924	0.898	0.720	0.941	0.864
MLP	0.957	0.872	0.940	0.961	0.996	0.984	0.966	0.131	0.961	0.850
E-D	0.801	0.796	0.991	0.988	0.995	0.947	0.938	-0.039	0.962	0.820

Table 8. Hyperparameter settings for the three models.

Hyperparam.	ModelS	ModelM	ModelL
Layers (L)	4	5	6
Hidden dim (d_{model})	256	256	256
FFN dim (d_{ff})	1024	1024	1024
Attn heads (h)	8	8	8
Dropout	0.2	0.2	0.2
Batch size	1024	1024	1024
Pretrain lr	8×10^{-4}	8×10^{-4}	8×10^{-4}
Finetune lr	1×10^{-4}	1×10^{-4}	1×10^{-4}
Weight decay	1×10^{-4}	1×10^{-4}	1×10^{-4}
ES patience (pretrain)	10	10	10
ES patience (finetune)	8	8	8

tion configuration fixed. Pretraining and fine-tuning share the same regularization and batch size, with a reduced learning rate and patience during fine-tuning to preserve small-signal features and adapt to long-tailed corners. Unlike the encoder-decoder setup used by Park *et al.* [7], our decoder-only BSIMAR avoids cross-attention, reducing compute/memory overhead and enabling a unified decoder to produce I-V, C-V, and Q-V responses with consistent physical relations.

- **Capacity scaling.** Depth increases from $L = 4 \rightarrow 5 \rightarrow 6$ (Model S/M/L), while width remains fixed (hidden dim $d_{\text{model}} = 256$, FFN dim $d_{\text{ff}} = 1024$, heads $h = 8$). This isolates depth as the capacity knob.
- **Regularization.** Dropout of 0.2 and weight decay of 10^{-4} are applied uniformly across S/M/L to curb overfitting in head-heavy regions without oversmoothing small-signal features.
- **Training schedule.** Batch size is 1024 for throughput and stable gradients. Pretraining uses a higher learning rate (8×10^{-4}) with early-stopping patience of 10; fine-tuning lowers the learning rate (1×10^{-4}) and patience (8) to adapt without drifting from the pretrain manifold.
- **Practical effect.** These settings yield monotonic accuracy gains with scale in common regions, while the decoder-only design and tail-aware training mitigate long-tail underfitting and improve cross-family consistency.

tency.

C. Circuit Simulation Integration

C.1. DC Analysis

C.2. Transition Analysis

D. Extended Study: Planar MOSFET

To assess portability beyond fin-based devices, we evaluate BSIMAR on planar MOSFETs (40 nm and 28 nm). Planar structures exhibit simpler electrostatics and charge partitioning than FinFETs; accordingly, under the same training budget, we expect faster convergence and lower error. The following tables report Mean Relative Error (MRE, %) and Coefficient of Determination (R^2) for zero-shot (pre-trained) and fine-tuned BSIMAR.

MRE trends. Fine-tuning consistently reduces average MRE across devices and nodes (Table 9): 40 nm nch_svt improves from 5.81% $\rightarrow$ 2.71%; 40 nm pch_svt from 7.01% $\rightarrow$ 3.92%; 28 nm nch_svt from 7.74% $\rightarrow$ 3.24%; and 28 nm pch_svt from 9.00% $\rightarrow$ 2.50%. Gains are strongest in derivative-sensitive quantities (e.g., Q_d , g_m). A few localized regressions (e.g., C_{gg} at 40 nm nch_svt, g_{ds} at 40 nm pch_svt) reflect sharper small-signal curvature; their impact is minor relative to overall averages.

R^2 corroboration. Correlation results (Table 10) mirror the MRE trends: 40 nm nch_svt average R^2 rises from 0.825 $\rightarrow$ 0.945, and 40 nm pch_svt from 0.741 $\rightarrow$ 0.969, with large improvements in previously weak small-signal metrics (e.g., C_{gd}). At 28 nm, already high baselines tighten further (to $\approx$ 0.968), indicating strong alignment across I-V, Q-V, and C-V families.

Taken together, BSIMAR works well on planar MOSFETs and, owing to their simpler physical model, achieves better results under comparable settings. The unified decoder with lightweight physical constraints exploits planar electrostatics to preserve cross-family consistency, while tail-aware training (smoothed loss and data filtering) quickly resolves small-signal curvature. Consequently, BSIMAR delivers lower errors and higher correlations on

Table 9. Comparison of BSIMAR’s Zero-Shot vs. Fine-Tuning performance, measured by Mean Relative Error (MRE, %), across 2 MOSFETs and 2 technologies. The first row presents the pre-training results. The second row (+ft) shows the MRE after fine-tuning. The values in color quantify the percentage change from zero-shot to fine-tuned performance.

Tech.	Device	I-V			Q-V			C-V			Average
		I_{ds}	g_m	g_{ds}	Q_g	Q_b	Q_d	C_{gg}	C_{gd}	C_{gs}	
40nm	nch_svt	6.64%	6.06%	12.20%	1.05%	1.38%	11.16%	0.39%	2.36%	7.06%	5.81%
	nch_svt+ft	0.89% -5.75%	2.04% -4.02%	10.88% -1.32%	0.90% -0.15%	0.87% -0.51%	2.83% -8.33%	1.51% +1.12%	1.93% -0.43%	2.52% -4.54%	2.71% -3.10%
	pch_svt	16.00%	4.19%	14.11%	0.59%	1.45%	13.64%	3.88%	2.63%	6.58%	7.01%
	pch_svt+ft	3.23% -12.77%	1.57% -2.62%	18.86% +4.75%	1.17% +0.58%	1.11% -0.34%	4.19% -9.45%	0.71% -3.17%	1.27% -1.36%	3.17% -3.41%	3.92% -3.09%
28nm	nch_svt	6.54%	7.99%	30.83%	1.87%	1.58%	10.77%	2.71%	1.80%	5.54%	7.74%
	nch_svt+ft	2.07% -4.47%	1.70% -6.29%	17.33% -13.50%	0.60% -1.27%	0.92% -0.66%	2.46% -8.31%	0.96% -1.75%	0.96% -0.84%	2.22% -3.32%	3.24% -4.50%
	pch_svt	6.64%	5.18%	40.68%	2.00%	1.18%	11.65%	3.08%	3.31%	7.24%	9.00%
	pch_svt+ft	1.91% -4.73%	2.58% -2.60%	8.98% -31.70%	0.61% -1.39%	0.44% -0.74%	2.83% -8.82%	0.84% -2.24%	1.73% -1.58%	2.58% -4.66%	2.50% -6.50%

Table 10. Comparison of BSIMAR’s Zero-Shot vs. Fine-Tuning performance, measured by the Coefficient of Determination (R^2), across 2 MOSFETs and 2 technologies. R^2 indicates a perfect correlation between predictions and the true data.

Tech.	Device	I-V			Q-V			C-V			Average
		I_{ds}	g_m	g_{ds}	Q_g	Q_b	Q_d	C_{gg}	C_{gd}	C_{gs}	
40nm	nch_svt	0.993	0.982	0.988	0.999	0.923	0.942	0.981	-0.379	0.994	0.825
	nch_svt+ft	0.999+0.006	1.000+0.018	0.998+0.010	1.000+0.001	0.968+0.045	0.997+0.054	0.997+0.017	0.178+0.558	0.998+0.005	0.904+0.079
	pch_svt	0.975	0.936	0.944	1.000	0.974	0.921	0.983	-1.056	0.994	0.741
	pch_svt+ft	0.995+0.020	0.999+0.063	0.978+0.034	0.999+0.001	0.986+0.012	0.996+0.075	1.000+0.016	0.554+1.610	1.000+0.005	0.945+0.204
28nm	nch_svt	0.988	0.961	0.967	0.999	0.955	0.962	0.998	0.835	1.000	0.963
	nch_svt+ft	0.997+0.009	0.999+0.038	0.778-0.189	1.000+0.001	0.993+0.038	0.997+0.035	0.955-0.043	0.955+0.120	1.000+0.000	0.969+0.006
	pch_svt	0.992	0.959	0.833	0.999	0.983	0.960	0.997	-0.021	0.999	0.856
	pch_svt+ft	0.999+0.007	0.998+0.039	0.988+0.155	1.000+0.001	0.998+0.015	0.999+0.039	1.000+0.002	0.734+0.755	1.000+0.001	0.968+0.113

planar devices with modest compute, reinforcing its generality and efficiency for compact modeling.

E. Advanced Loss Comparison: MAE, MSE, LDS-MAE

Table 11 summarizes fine-tuning results for compact modeling of 7 nm *n*- and *p*-channel svt MOSFETs, reporting MRE (%) across nine characteristics: drain current (I_{ds}), transconductance (g_m), output conductance (g_{ds}), gate/bulk/drain charges (Q_g , Q_b , Q_d), and intrinsic capacitances (C_{gg} , C_{gd} , C_{gs}). Reading the table, **MAE consistently outperforms MSE on overall accuracy**: it reduces the average MRE from 7.38 % to 5.32 % on nch_svt and from 11.44 % to 6.78 % on pch_svt. While MSE remains competitive on a few capacitances (e.g., C_{gg} , C_{gd}), MAE generally yields lower errors across the majority of targets.

MAE + LDS further improves characteristics that suffer larger errors—most notably g_{ds} (12.81 % vs. 19.12%–21.61 % on nch_svt; 3.75 % vs. 12.84 % on pch_svt) and several charge-related quantities—while *slightly* sacrificing precision on some easier characteristics (e.g., C_{gg} , C_{gd} and occasionally I_{ds} on pch_svt). This targeted regularization delivers the best overall averages: **3.84 %** on nch_svt and **6.29 %** on pch_svt, showing great potential for improving overall accuracy in physics-guided compact models.

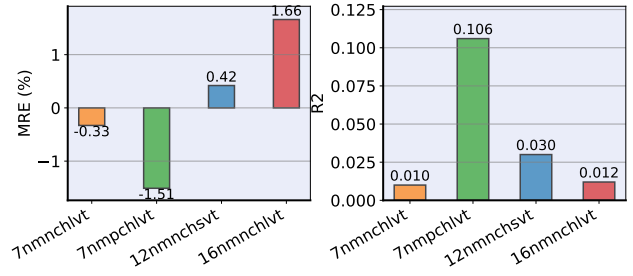


Figure 9. Impact of filtering on MRE and R^2 across devices. Negative Δ MRE indicates error reduction; positive Δ R^2 indicates stronger correlation.

F. Data Filtering Strategy

As shown in Figure 9, filtering has heterogeneous effects across devices. For 7 nm nchlvt and 7 nm pchlvt, MRE decreases by -0.33% and -1.51% , respectively; for 12 nm nchsvt and 16 nm nchlvt, MRE increases by $+0.42\%$ and $+1.66\%$. Correlation remains modest overall: $R^2 = 0.106$ for 7 nm pchlvt (highest), and $R^2 \in \{0.010, 0.030, 0.012\}$ for the others. The mismatch between Δ MRE and Δ R^2 indicates that error magnitude and pointwise correlation capture complementary aspects of fit; we therefore report both metrics and emphasize cross-family consistency in downstream evaluations. Importantly, not all devices across all technologies benefit from data filtering during fine-tuning; in some cases error increases and correlation changes are negligible. We treat

Table 11. LDS-MAE loss ablation at the **fine-tuning** stage: Mean Relative Error (MRE, %) across BSIMAR predictions for n -channel and p -channel SVT devices at 7nm. Results compare training objectives: MSE, MAE, and MAE+LDS.

Loss	I_{ds}	g_m	g_{ds}	Q_g	Q_b	Q_d	C_{gg}	C_{gd}	C_{gs}	Average
MSE	2.14%	3.13%	21.61%	5.72%	13.04%	13.04%	1.72%	1.57%	4.48%	7.38%
MAE	2.17%	2.97%	19.12%	6.08%	2.86%	4.90%	3.06%	1.89%	4.79%	5.32%
MAE+LDS	1.32%	1.61%	12.81%	3.51%	5.02%	4.15%	1.79%	1.89%	2.49%	3.84%
MSE	11.68%	12.59%	16.87%	10.82%	11.63%	17.96%	3.05%	7.29%	11.10%	11.44%
MAE	5.78%	11.83%	12.84%	11.22%	3.64%	5.57%	2.83%	1.62%	5.66%	6.78%
MAE+LDS	6.95%	10.83%	3.75%	8.30%	3.76%	7.16%	4.10%	4.14%	7.58%	6.29%

756 filtering as a device–and technology–specific intervention.
757 Balancing the pre–training and fine–tuning datasets (e.g.,
758 reweighting tails and harmonizing per–technology distribu-
759 tions) will be addressed in future work.